CODE FOR ARDUINO UNO

```
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 2, 16);
int sensorPin = A0;
void setup()
{
 Serial.begin(9600);
 lcd.begin(12,2,6*5);
 pinMode(2, OUTPUT);
 pinMode(3, OUTPUT);
 pinMode(4, OUTPUT);
}
void loop() {
 int sensorValue = analogRead(sensorPin);
 Serial.println(sensorValue);
 int turbidity = map(sensorValue, 0, 750, 100, 0);
 delay(100);
 lcd.setCursor(0, 0);
 lcd.print("turbidity:");
 lcd.print("   ");
 lcd.setCursor(10, 0);
 lcd.print(turbidity);
 delay(100);
 if (turbidity < 20) {
  digitalWrite(2, HIGH);
  digitalWrite(3, LOW);
  digitalWrite(4, LOW);
  lcd.setCursor(0, 1);
```

```
  lcd.print(" its CLEAR ");
 }
 if ((turbidity > 20) && (turbidity < 50)) {
  digitalWrite(2, LOW);
  digitalWrite(3, HIGH);
  digitalWrite(4, LOW);
  lcd.setCursor(0, 1);
  lcd.print(" its CLOUDY ");
 }
 if (turbidity > 50) {
  digitalWrite(2, LOW);
  digitalWrite(3, LOW);
  digitalWrite(4, HIGH);
  lcd.setCursor(0, 1);
  lcd.print(" its DIRTY ");
 }
}
```